TechSolutions
SARL

Network Infrastructure Architecture Report

Tech Solutions (SARL)

Version 1.0 — January 7, 2026

Executive Summary

This document provides a complete professional overview of the network infrastructure designed for Tech Solutions (SARL). The architecture simulates a modern enterprise network using GNS3, featuring routed backbone segments, VLAN segmentation, firewall policies, and Linux-based services.

The objective is to model a scalable, secure, and realistic environment for engineering, training, and testing purposes. It provides system engineers and network architects with a clear understanding of the network's logical and physical design, implementation steps, and operational considerations.

Contents

1 Introduction

The Tech Solutions (SARL) network architecture project is a fully simulated enterprise environment developed within GNS3. It includes multi-router infrastructure, departmental VLANs, dynamic routing, and server resources.

The purpose of this report is to present:

- network design principles,
- device roles and interconnections,
- used technologies and protocols,
- deployment and implementation steps,
- configuration and testing methodology.

1.1 Departmental Allocation and Logical Topology

After the integration of the Internet service, the ISP allocated the address space **172.24.0.0/14** to the enterprise. The logical topology and IP allocation for each department are summarized below:

Department	Local Router	Service / VM	PC Clients	Role in the Enterprise
Web / Marketing	RZ-1	Web Server	8905	Manages the website and customer-facing interactions
Supervision / IT	RZ-2	Monitoring Server	1465	Monitors server and network performance
Base de données / Gestion	RZ-3	Database Server	489	Centralizes and secures client and internal data
Partage / Collaboration	RZ-4	NFS Server	165	Enables internal file and document sharing

1.2 Subnetting (VLSM) Diagram

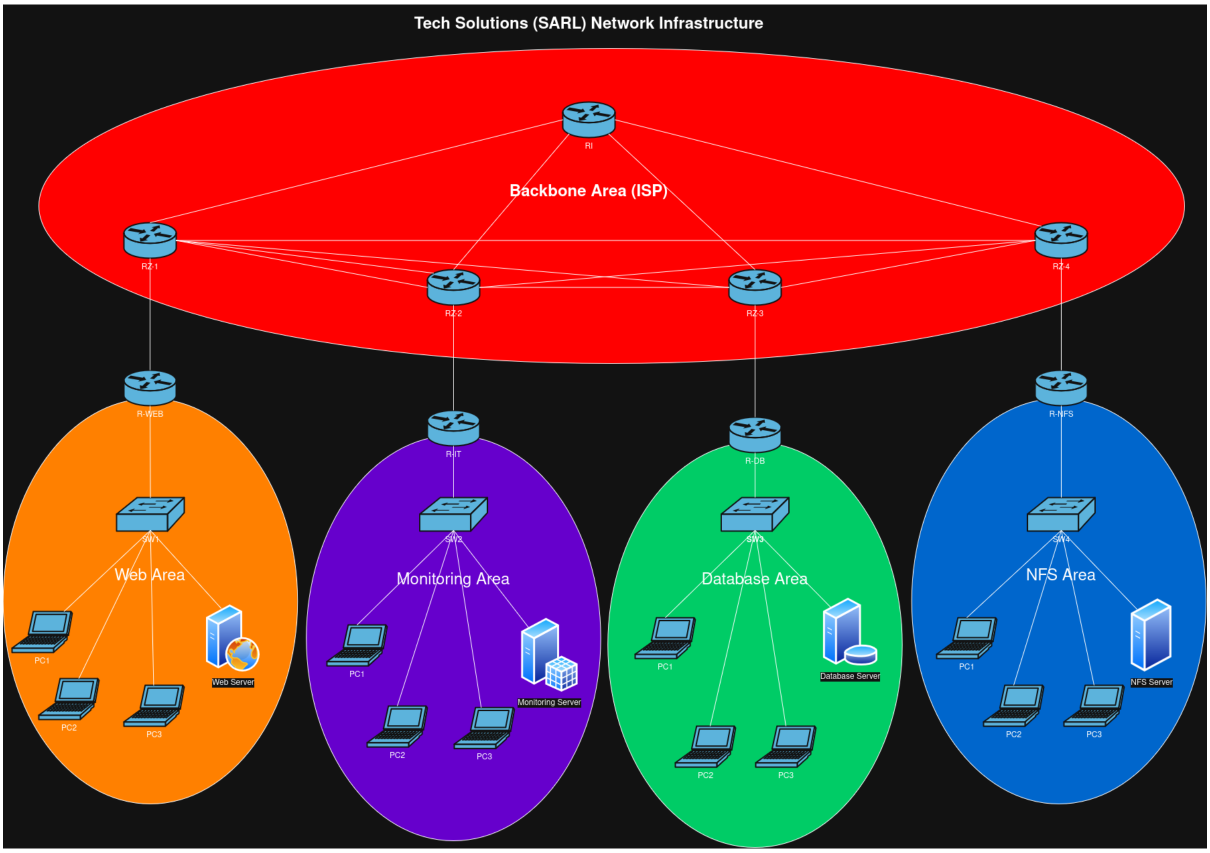
The network 172.24.0.0/14 has 262142 hosts.
Your subnets need 11032 hosts.

NAME	HOSTS NEEDED	HOSTS AVAILABLE	UNUSED HOSTS	NETWORK ADDRESS	SLASH	MASK	USABLE RANGE	BROADCAST
Web	8907	16382	7475	172.24.0.0	/18	255.255.192.0	172.24.0.1 - 172.24.63.254	172.24.63.255
IT	1467	2046	579	172.24.64.0	/21	255.255.248.0	172.24.64.1 - 172.24.71.254	172.24.71.255
DB	491	510	19	172.24.72.0	/23	255.255.254.0	172.24.72.1 - 172.24.73.254	172.24.73.255
NFS	167	254	87	172.24.74.0	/24	255.255.255.0	172.24.74.1 - 172.24.74.254	172.24.74.255

2 Technologies Used

- **GNS3 3.0.5** — Network simulation environment.
- **Cisco Routers (3725/3745)** — Core routing infrastructure.
- **Ubuntu Server 20.04** — Server and network service hosting.
- **OSPF** — Dynamic routing protocol.
- **VLANs** — Logical segmentation and broadcast containment.
- **Nmap & Wireshark** — Analysis and testing tools.
- **ZeroTier** — Potential remote access overlay network.
- **SSH / Telnet** — Device management protocols.

3 Architecture Design



3.1 Core Layer

The backbone contains the primary routing equipment enabling communication between all segments.

3.2 Distribution Layer

VLAN segmentation and policy enforcement occur at this layer.

3.3 Access Layer

End-user devices and servers connect at this level.

4 Implementation Steps

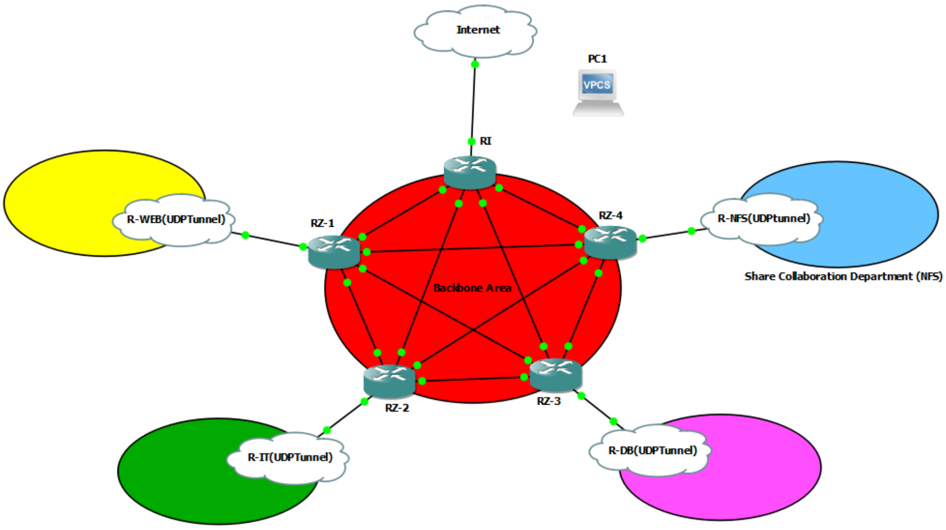
1. Prepare GNS3 environment and import appliance templates.
2. Deploy routers, switches, and servers.
3. Configure VLANs and interfaces.
4. Configure OSPF across backbone routers.
5. Implement ACLs and security policies.
6. Verify connectivity and routing.
7. Test bandwidth, latency, and redundancy.

5 Security Model

Security policies were applied on multiple layers:

- ACLs restricting cross-department traffic.
- VLAN isolation.
- SSH-enabled secure device management.
- Logging and monitoring using Linux servers.

6 Backbone Infrastructure



6.1 Purpose

The backbone provides high-availability routed connectivity between all departments, the Internet, and security services. It ensures redundancy, dynamic routing, and controlled external access.

6.2 Architecture Overview

The backbone consists of:

- Four core routers (CR1–CR4)
- OSPF dynamic routing
- FortiGate firewall (VPN + filtering)
- Internet edge router with NAT

6.3 OSPF Design

- OSPF Area 0 (Backbone)
- All core routers participate in Area 0
- Point-to-point links between core routers
- Route summarization applied toward distribution routers

6.4 VPN and Security

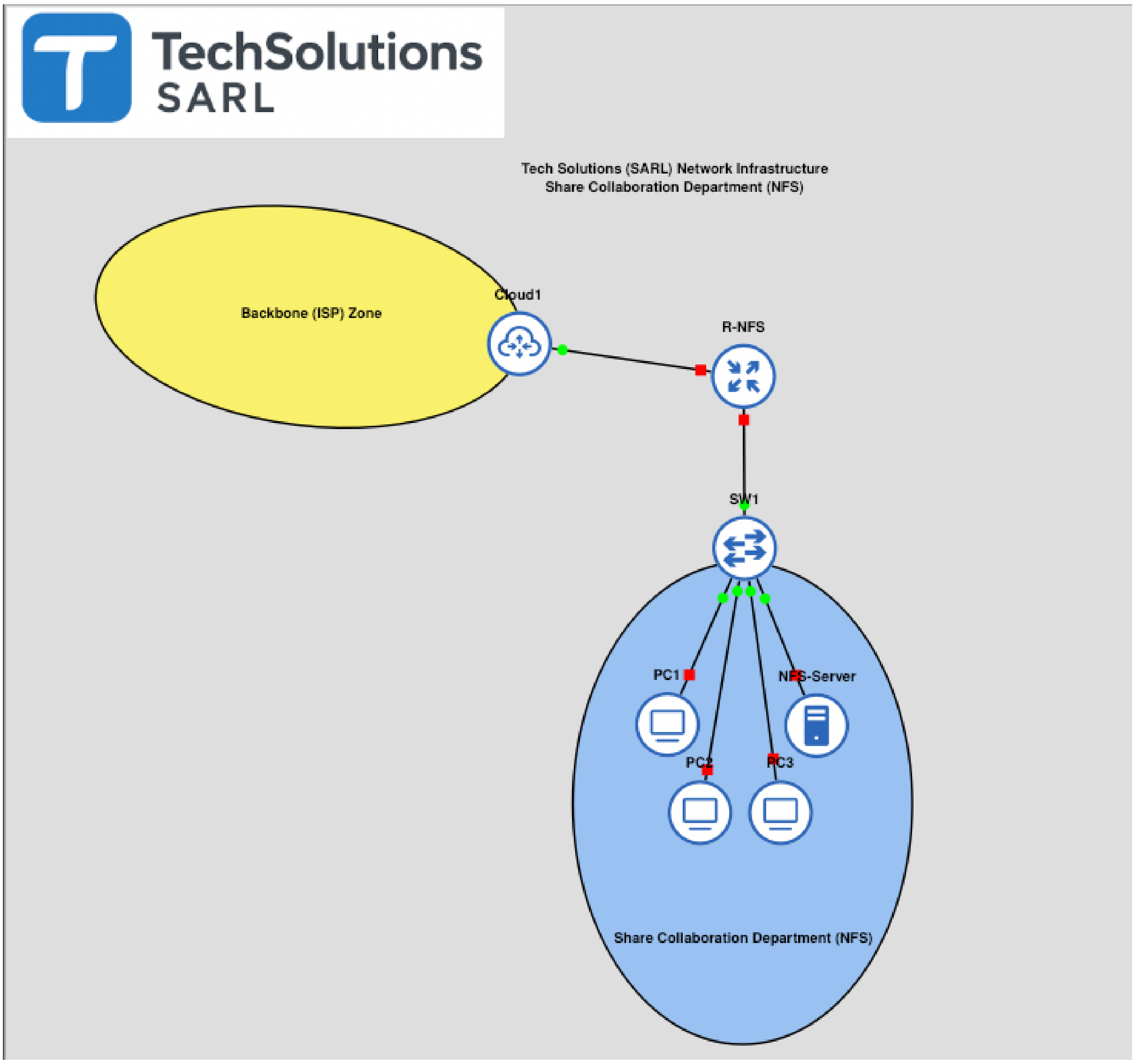
- FortiGate provides site-to-site and remote-access VPN

- Encrypted tunnels secure management and inter-site traffic
- Firewall policies restrict inbound/outbound traffic

6.5 Internet Router (NAT)

- Static default route toward ISP
- PAT (NAT overload) for internal subnets
- Controlled port forwarding for public services

7 NFS Department



7.1 Purpose

This department provides centralized file sharing services for internal collaboration and data exchange.

7.2 Architecture

- Linux NFS Server (Ubuntu Server)
- Client workstations
- Dedicated router and access switch

7.3 Server Configuration Overview

- NFS service enabled
- Shared directories defined in `/etc/exports`
- Access restricted to internal subnets

7.4 Client Configuration Script Overview

Clients mount shared directories automatically at boot using:

- `mount -t nfs`
- `/etc/fstab`

7.5 Router Script Overview

- Interface IP configuration
- OSPF advertisement
- ACL allowing NFS traffic only

7.6 Switch Script Overview

- VLAN assignment
- Access ports for clients
- Trunk port toward router

7.7 Scripts and Configuration

7.7.1 NFS Router Configuration

```
! ZouariOmar <zouariomar20@gmail.com>
! Since:         2025-11-19
! LastUpdate:    2026-01-05
! RouterName:    R-nfs
! RouterVersion: C3745
! COPY-THAN-PASTE

conf t
hostname R-nfs

! ============================
```

```
! ===== INTERFACES PART ======
! === F0/1(WAN) - F0/0(LAN) ==
! ============================

! === FastEthernet0/1 (WAN) ===
interface fa0/1
  description R-nfs to UDP Tunnel
  ip address 10.0.0.2 255.255.255.0
  no shutdown
  exit

! === FastEthernet0/0 (LAN) ===
interface fa0/0
  description R-nfs to SW1
  ip address 172.24.74.1 255.255.255.0
  no shutdown
  exit

! ============================
! ====== SERVICES PART =======
! ==== OSPF - DHCP - DNS =====
! ============================

! === OSPF ===
router ospf 1
  router-id 7.7.7.7
  network 10.0.0.0 0.0.0.255 area 0
  network 172.24.74.0 0.0.0.255 area 0
  passive-interface default
  no passive-interface FastEthernet0/1
  ! NOTE: ‘f0/0‘ should not be no ‘passive-interface‘
  default-information originate
  exit

! Default route toward backbone
ip route 0.0.0.0 0.0.0.0 10.0.0.1

! === DHCP ===
ip dhcp pool NFS-DHCP
  network 172.24.74.0 255.255.255.0
  default-router 172.24.74.1
  dns-server 8.8.8.8
  exit

! Exclude @GW
ip dhcp excluded-address 172.24.74.1

! Exclude @NFS-server
ip dhcp excluded-address 172.24.74.2

! === DNS ===
```

```
ip domain-lookup
  ip name-server 8.8.8.8
  exit

! ==================================
! ========= SECURITY PART ==========
! == CONSOLE - SSH - TELNET - ACL ==
! ==================================

! Create an Admin user
username admin privilege 15 secret <STRONG_PASSWORD>

! Encrypt all passwords
service password-encryption

! === Console ===
line console 0
  login local
  exec-timeout 5 0    ! Auto-logout after 5min
  logging synchronous ! Clean console display
  exit

! === SSH ===
crypto key generate rsa usage-keys label SSH modulus 2048
ip ssh version 2
! ip access-list standard SSH_ONLY
!   permit X.X.X.X X.X.X.X
!   deny any
line vty 0 4
  login local
  transport input ssh
! access-class SSH_ONLY in
  exec-timeout 5 0
  logging synchronous
  exit

! === Telnet ===
! ip access-list standard TELNET_ONLY
!   permit 10.0.0.0 0.0.0.255
!   deny any
line vty 5 15
  login local
  transport input telnet
! access-class TELNET_ONLY in
  exec-timeout 5 0
  logging synchronous
  exit

! Disable unused services
! no ip http server          ! HTTP
! no ip http secure-server   ! HTTPS
```

```
! no cdp run                    ! CDP

write

! ========================
! === VERIFICATION PART ===
! ========================

! === OSPF Interfaces ===
! Check which interfaces run OSPF and their state
show ip ospf interface brief
show ip ospf interface
show ip protocols

! === OSPF Interfaces ===
! Verify adjacency with other routers (if backbone is up)
show ip ospf neighbor
show ip ospf neighbor detail

! === OSPF Routes ===
! Verify OSPF learned routes
show ip route ospf
show ip route

! === Default Route / Redistribution ===
! Check default route advertisement
show ip route 0.0.0.0
show running-config | include default-information

! === LAN Connectivity / DHCP ===
! Verify DHCP bindings and pool
show ip dhcp binding
show ip dhcp pool

! === DNS Resolution ===
! Verify DNS lookup from router
ping 8.8.8.8
ping www.google.com
```

Listing 1: NFS Router Configuration

7.7.2 NFS Server Script

```
#!/usr/bin/env bash
# ========================================
# .FILE
#   nfs-server-conf
#
# .SYNOPSIS
#   Bash script to configure NFS-Linux server
#
```

```
# .DESCRIPTION
#   I write this script for
#   ‘Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
    PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
    /Linux‘
#   (ubuntu-24.04.3-live-server-amd64)
#
# .PARAMETER NONE
#   NONE
#
# .INPUTS
#   NONE
#
# .OUTPUTS
#   NONE
#
# .NOTES
#   Version        : 1.0
#   Author         : @ZouariOmar <zouariomar20@gmail.com>
#   Created        : 11/22/2025
#   Change Log     : 9780b36
#
# .EXAMPLE
#   > ./nfs-server-conf
#   --
#   > NET_INTERFACE="enp2s0"       \
#     NET_ADDRESS="172.24.74.2/24" \
#     NET_GW="172.24.74.1"         \
#     NET_DNS_SERVERS="[8.8.8.8]"  \
#     SHARED_DIR='/srv/nfs/share' \
#     NFS_GROUP='zouari_omar'
#     ./nfs_server_conf
# ========================================

#################################
### CUSTOM CONFIGURATION PART ###
#################################
NET_INTERFACE="enp2s0"
NET_SUBNET='X.X.X.X/XX'
NET_ADDRESS="X.X.X.X/XX"
NET_GW="X.X.X.X"
NET_DNS_SERVERS="[X.X.X.X]"
SHARED_DIR='path_to_shared_directory'
NFS_GROUP='group_to_chown_shared_dir'

##################################
### GENERAL CONFIGURATION PART ###
##################################

# Set keymap into fr
```

```
sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
    keyboard

# Update the system
sudo apt update -y && sudo apt upgrade -y

# Change the current hostname
hostnamectl set-hostname nfs-server

##################################
### NETWORK CONFIGURATION PART ###
##################################

# Set the private static ip address
sudo bash -c "cat << 'EOF' > /etc/netplan/01-static.yaml
network:
  version: 2
  renderer: networkd
  ethernets:
    ${NET_INTERFACE}:
      dhcp4: no
      addresses:
        - ${NET_ADDRESS}
      nameservers:
        addresses: ${NET_DNS_SERVERS}
      routes:
        - to: default
          via: ${NET_GW}
EOF"

# Change 01-static.yaml accessibility
sudo chmod 600 /etc/netplan/01-static.yaml

# Apply changes
sudo netplan apply

#####################################
### NFS-SERVER CONFIGURATION PART ###
#####################################

# Install Server NFS-SERVER package
sudo apt install -y nfs-kernel-server nfs-common

# Enable NFS service
sudo systemctl enable --now nfs-server

# Create and Export a Shared Directory
sudo mkdir -p ${SHARED_DIR}

# Add local subnet to the ACL (Access Control List)
```

```
echo "${SHARED_DIR} ${NET_SUBNET}(rw,sync,no_subtree_check)" |
   sudo tee -a /etc/exports

# Apply exports
sudo exportfs -ra

# Give grp permission
# NOTE: To add new group `sudo groupadd NFS_GROUP`
# NOTE: To delete group `sudo groupdel NFS_GROUP`
sudo chown -R :${NFS_GROUP} ${SHARED_DIR}
sudo chmod -R 775 ${SHARED_DIR}

# Configure Firewall
sudo ufw allow from ${NET_SUBNET} to any port nfs
```

Listing 2: NFS Server Configuration Script

7.7.3 NFS Client Script

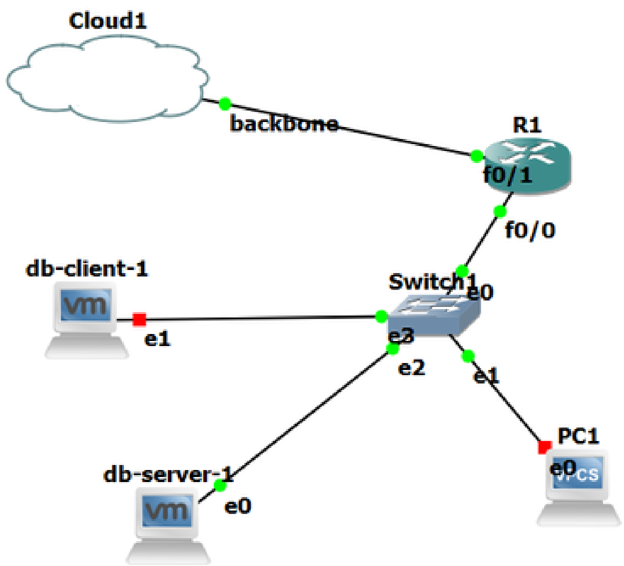
```
#!/usr/bin/env bash
# ========================================
# .FILE
#   nfs_client_conf
#
# .SYNOPSIS
#   Bash script to configure NFS-Linux client
#
# .DESCRIPTION
#   I write this script for
#   `Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
   PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
   /Linux`
#   (ubuntu-24.04.3-live-server-amd64)
#
# .PARAMETER NONE
#   NONE
#
# .INPUTS
#   NONE
#
# .OUTPUTS
#   NONE
#
# .NOTES
#   Version       : 1.0
#   Author        : @ZouariOmar <zouariomar20@gmail.com>
#   Created       : 11/22/2025
#   Change Log    : 9780b36
#
# .EXAMPLE
#   > ./nfs-client-conf
```

```
#    --
#    > NET_INTERFACE=enp2s0          \
#      SERVER_ADDRESS='172.24.74.2' \
#      SHARED_DIR='/srv/nfs/share'  \
#      MOUNT_POINT='/mnt/nfs'
#      ./nfs_client_conf
# =========================================

#################################
### CUSTOM CONFIGURATION PART ###
#################################
NET_INTERFACE="enp2s0"
SERVER_ADDRESS="X.X.X.X"
SHARED_DIR='path_to_shared_directory'
MOUNT_POINT='path_to_mount_directory'

##################################
### GENERAL CONFIGURATION PART ###
##################################

# Set keymap into fr
sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
   keyboard

# Update the system
sudo apt update -y && sudo apt upgrade -y

# Change the current hostname
hostnamectl set-hostname nfs-client

##################################
### NETWORK CONFIGURATION PART ###
##################################

# Set dhcp ip address
sudo bash -c "cat << 'EOF' > /etc/netplan/01-enp2s0.yaml
network:
  version: 2
  ethernets:
    ${NET_INTERFACE}:
      dhcp4: true
EOF"

# Change 01-enp2s0.yaml accessibility
sudo chmod 600 /etc/netplan/01-enp2s0.yaml

# Apply changes
sudo netplan apply

#####################################
### NFS-CLIENT CONFIGURATION PART ###
```

```
####################################

# INFO: To change group name: ‘sudo groupmod -n new-name current-
    name‘
# Install NFS-CLIENT package
sudo apt install nfs-common

# Create mount point
sudo mkdir -p ${MOUNT_POINT}

# Mount the share
sudo mount ${SERVER_ADDRESS}:${SHARED_DIR} ${MOUNT_POINT}

# Verify mount
df -h | grep nfs

# Umount the share
# sudo umount ${MOUNT_POINT}

# Mount NFS share safely using systemd automount; system boots
    normally even if the NFS server is down.
echo "${SERVER_ADDRESS}:${SHARED_DIR}  ${MOUNT_POINT}  nfs
    defaults,_netdev,x-systemd.automount,nofail  0  0" | sudo tee
    -a /etc/fstab >/dev/null
```

Listing 3: NFS Client Configuration Script

8 Database Department



8.1 Purpose

This department hosts the enterprise MySQL database, centralizing business and client data securely.

8.2 Architecture

- Linux Database Server
- MySQL service
- Restricted client access

8.3 MySQL Configuration

- MySQL bound to internal interface only
- Dedicated database users per service
- Regular backups scheduled

8.4 Security

- ACLs limiting database access
- Strong authentication
- Network isolation via VLAN

8.5 Scripts and Configuration

8.5.1 Database Router Configuration

```
! ZouariOmar <zouariomar20@gmail.com>
! Since:         2025-12-23
! LastUpdate:    2026-01-05
! RouterName:    R-db
! RouterVersion: C3745
! COPY-THAN-PASTE

conf t
hostname R-db

! ===========================
! ===== INTERFACES PART ======
! === F0/1(WAN) - F0/0(LAN) ==
! ===========================

! === FastEthernet0/1 (WAN) ===
interface FastEthernet0/1
  description R-db to UDP Tunnel / Backbone
  ip address 10.1.0.2 255.255.255.252
  no shutdown
  exit
```

```

! === FastEthernet0/0 (LAN) ===
interface FastEthernet0/0
  description R-db to DB-LAN
  ip address 172.24.72.1 255.255.254.0
  no shutdown
  exit

! ============================
! ====== SERVICES PART =======
! ==== OSPF - DHCP - DNS =====
! ============================

! === OSPF ===
router ospf 1
  router-id 6.6.6.6
  log-adjacency-changes
  network 10.1.0.0 0.0.0.3 area 0
  network 172.24.72.0 0.0.1.255 area 0
  passive-interface default
  no passive-interface FastEthernet0/1
  default-information originate
  exit

! Default route toward backbone
ip route 0.0.0.0 0.0.0.0 10.1.0.1

! === DHCP ===
! Excluded addresses
ip dhcp excluded-address 172.24.72.1 172.24.72.2

ip dhcp pool DB-LAN
  network 172.24.72.0 255.255.254.0
  default-router 172.24.72.1
  dns-server 8.8.8.8
  exit

! === DNS ===
ip domain-lookup
ip name-server 8.8.8.8
ip name-server 1.1.1.1

! ============================
! ====== SECURITY PART =======
! CONSOLE - SSH - TELNET - ACL
! ============================

! Create Admin user
username admin privilege 15 secret <STRONG_PASSWORD>

! Encrypt all passwords
```

```
service password-encryption

! === Console ===
line console 0
  login local
  exec-timeout 5 0
  logging synchronous
  exit

! === SSH ===
crypto key generate rsa usage-keys label SSH modulus 2048
ip ssh version 2

line vty 0 4
  login local
  transport input ssh
  exec-timeout 5 0
  logging synchronous
  exit

! === Telnet (optional / legacy) ===
line vty 5 15
  login local
  transport input telnet
  exec-timeout 5 0
  logging synchronous
  end

! Disable unused services
no ip http server
no ip http secure-server
! no cdp run

write

! ========================
! === VERIFICATION PART ===
! ========================

! === OSPF Interfaces ===
! Check which interfaces run OSPF and their state
show ip ospf interface brief
show ip ospf interface
show ip protocols

! === OSPF Interfaces ===
! Verify adjacency with other routers (if backbone is up)
show ip ospf neighbor
show ip ospf neighbor detail

! === OSPF Routes ===
```

```
! Verify OSPF learned routes
show ip route ospf
show ip route

! === Default Route / Redistribution ===
! Check default route advertisement
show ip route 0.0.0.0
show running-config | include default-information

! === LAN Connectivity / DHCP ===
! Verify DHCP bindings and pool
show ip dhcp binding
show ip dhcp pool

! === DNS Resolution ===
! Verify DNS lookup from router
ping 8.8.8.8
ping www.google.com
```

Listing 4: DB Router Configuration

8.5.2 Database Server Script

```
#!/usr/bin/env bash
# ==========================================
# .FILE
#   db_server_conf
#
# .SYNOPSIS
#   Bash script to configure DB-Linux server
#
# .DESCRIPTION
#   I write this script for
#   ‘Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP
   PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU
   /Linux‘
#   (ubuntu-24.04.3-live-server-amd64)
#
# .PARAMETER NONE
#   NONE
#
# .INPUTS
#   NONE
#
# .OUTPUTS
#   NONE
#
# .NOTES
#   Version        : 1.0
#   Author         : @ZouariOmar <zouariomar20@gmail.com>
#   Created        : 11/22/2025
```

```
#   Change Log    : 9780b36
#
# .EXAMPLE
#   > ./db_server_conf
#   --
#   > NET_INTERFACE="enp2s0"                     \
#     NET_ADDRESS="172.24.74.2/24"               \
#     NET_GW="172.24.74.1"                       \
#     NET_DNS_SERVERS="[8.8.8.8]"                \
#     DB_ACCESS_CLIENT="clientdb'@'192.168.2.%'"
#     ./db_server_conf
# =========================================

#################################
### CUSTOM CONFIGURATION PART ###
#################################
NET_INTERFACE="enp2s0"
NET_ADDRESS="X.X.X.X/XX"
NET_GW="X.X.X.X"
NET_DNS_SERVERS="[X.X.X.X]"
DB_ACCESS_CLIENT="'db_client'@'X.X.X.X'"

##################################
### GENERAL CONFIGURATION PART ###
##################################

# Set keymap into fr
sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
   keyboard

# Update the system
sudo apt update -y && sudo apt upgrade -y

# Change the current hostname
hostnamectl set-hostname nfs-server

##################################
### NETWORK CONFIGURATION PART ###
##################################

# Set the private static ip address
sudo bash -c "cat << 'EOF' > /etc/netplan/01-static.yaml
network:
  version: 2
  renderer: networkd
  ethernets:
    ${NET_INTERFACE}:
      dhcp4: no
      addresses:
        - ${NET_ADDRESS}
      nameservers:
```

```
        addresses: ${NET_DNS_SERVERS}
      routes:
        - to: default
          via: ${NET_GW}
EOF"

# Change 01-static.yaml accessibility
sudo chmod 600 /etc/netplan/01-static.yaml

# Apply changes
sudo netplan apply

#####################################
### DB-SERVER CONFIGURATION PART ###
#####################################

# Install MySQL Server
sudo apt install mysql-server -y

# Enable mysql service
sudo systemctl enable --now mysql

# Allow Remote Connections
sudo sed -i 's/^bind-address\s*=.*/bind-address = 0.0.0.0/' /etc/
   mysql/mysql.conf.d/mysqld.cnf
sudo systemctl restart mysql

# Firewall and Port Check
sudo ufw allow 3306/tcp
sudo ufw reload

# Database and User Creation
sudo mysql -e "CREATE DATABASE IF NOT EXISTS labdb; CREATE USER
   IF NOT EXISTS ${DB_ACCESS_CLIENT} IDENTIFIED BY 'StrongPass123
   !'; GRANT ALL PRIVILEGES ON labdb.* TO ${DB_ACCESS_CLIENT};
   FLUSH PRIVILEGES;"

# Create Table and Insert Data
sudo mysql -e "USE labdb; CREATE TABLE IF NOT EXISTS users (id
   INT AUTO_INCREMENT PRIMARY KEY, username VARCHAR(50), email
   VARCHAR(100)); INSERT INTO users (username, email) VALUES ('
   server_user1','srv1@mail.com'),('server_user2','srv2@mail.com
   '); SELECT * FROM users;"
```

Listing 5: DB Server Configuration Script

8.5.3 Database Client Script

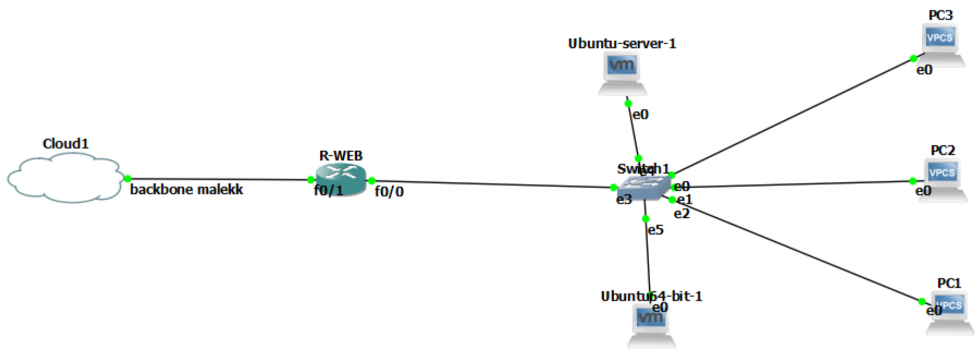
```
#!/usr/bin/env bash
# ========================================
# .FILE
```

```
#   db_client_conf
#
# .SYNOPSIS
#   Bash script to configure DB-Linux client
#
# .DESCRIPTION
#   I write this script for
#   ‘Linux nfs-server 6.8.0-88-generic #89-Ubuntu SMP PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 GNU/Linux‘
#   (ubuntu-24.04.3-live-server-amd64)
#
# .PARAMETER NONE
#   NONE
#
# .INPUTS
#   NONE
#
# .OUTPUTS
#   NONE
#
# .NOTES
#   Version       : 1.0
#   Author        : @ZouariOmar <zouariomar20@gmail.com>
#   Created       : 11/22/2025
#   Change Log    : 9780b36
#
# .EXAMPLE
#   > ./nfs_client_conf
#   --
#   > NET_INTERFACE=enp2s0 \
#     DB_SERVER_IP_ADDRESS="172.24.72.2" \
#     DB_CLIENT_NAME="client" \
#     DB_CLIENT_PASSWORD="client123"
#     ./db_client_conf
# =========================================

#################################
### CUSTOM CONFIGURATION PART ###
#################################
NET_INTERFACE="enp2s0"
DB_SERVER_IP_ADDRESS="X.X.X.X"
DB_CLIENT_NAME="client_name"
DB_CLIENT_PASSWORD="client_password"
HOSTNAME="db-client"

##################################
### GENERAL CONFIGURATION PART ###
##################################

# Set keymap into fr
```

```
sudo sed -i 's/XKBLAYOUT=".*"/XKBLAYOUT="fr"/' /etc/default/
    keyboard

# Update the system
sudo apt update -y && sudo apt upgrade -y

# Change the current hostname
hostnamectl set-hostname ${HOSTNAME}

##################################
### NETWORK CONFIGURATION PART ###
##################################

# Set dhcp ip address
sudo bash -c "cat << 'EOF' > /etc/netplan/01-enp2s0.yaml
network:
  version: 2
  ethernets:
    ${NET_INTERFACE}:
      dhcp4: true
EOF"

# Change 01-enp2s0.yaml accessibility
sudo chmod 600 /etc/netplan/01-enp2s0.yaml

# Apply changes
sudo netplan apply

####################################
### DB-CLIENT CONFIGURATION PART ###
####################################

# Install MySQL Client
sudo apt install mysql-client -y

# Remote Connection && Retrieve Server Data
mysql -h ${DB_SERVER_IP_ADDRESS} -u ${DB_CLIENT_NAME} -p -e "SHOW
    DATABASES; USE labdb; SHOW TABLES; SELECT * FROM users;"
```

Listing 6: DB Client Configuration Script

9 Web Department



9.1 Purpose

Hosts the public-facing enterprise website and internal web services.

9.2 Architecture

- Linux Web Server
- Apache2 HTTP server
- Client workstations

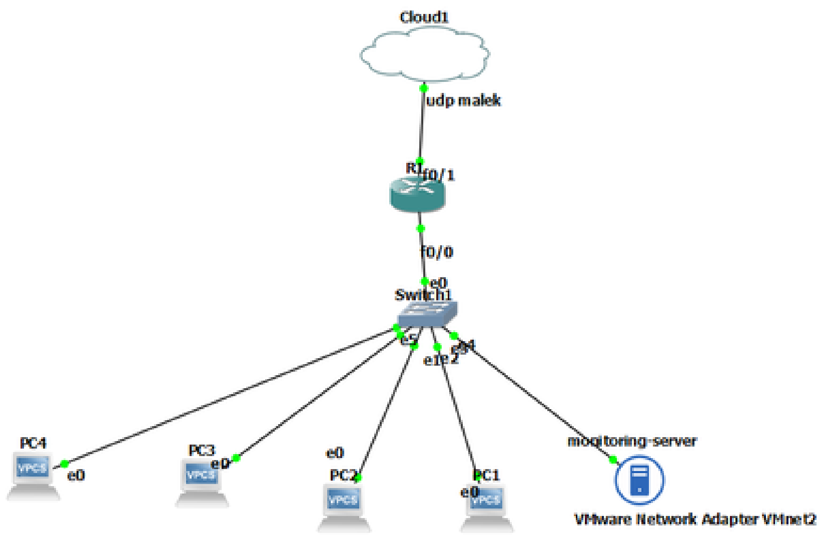
9.3 Apache2 Configuration

- Virtual hosts enabled
- Static and dynamic content support
- Logs stored for auditing

9.4 Router and Security

- Port forwarding via NAT
- ACLs permitting HTTP/HTTPS
- OSPF routing integration

10 Monitoring Department



10.1 Purpose

Provides real-time monitoring, metrics collection, and visualization of the entire infrastructure.

10.2 Architecture

- Linux Monitoring Server
- Prometheus for metrics collection
- Grafana for visualization

10.3 Prometheus

- Scrapes routers and servers
- Node exporter installed on Linux hosts
- Custom alert rules defined

10.4 Grafana

- Dashboards for CPU, RAM, network usage
- Role-based access control
- Historical data analysis

10.5 Screenshots

10.5.1 Prometheus Web Interface

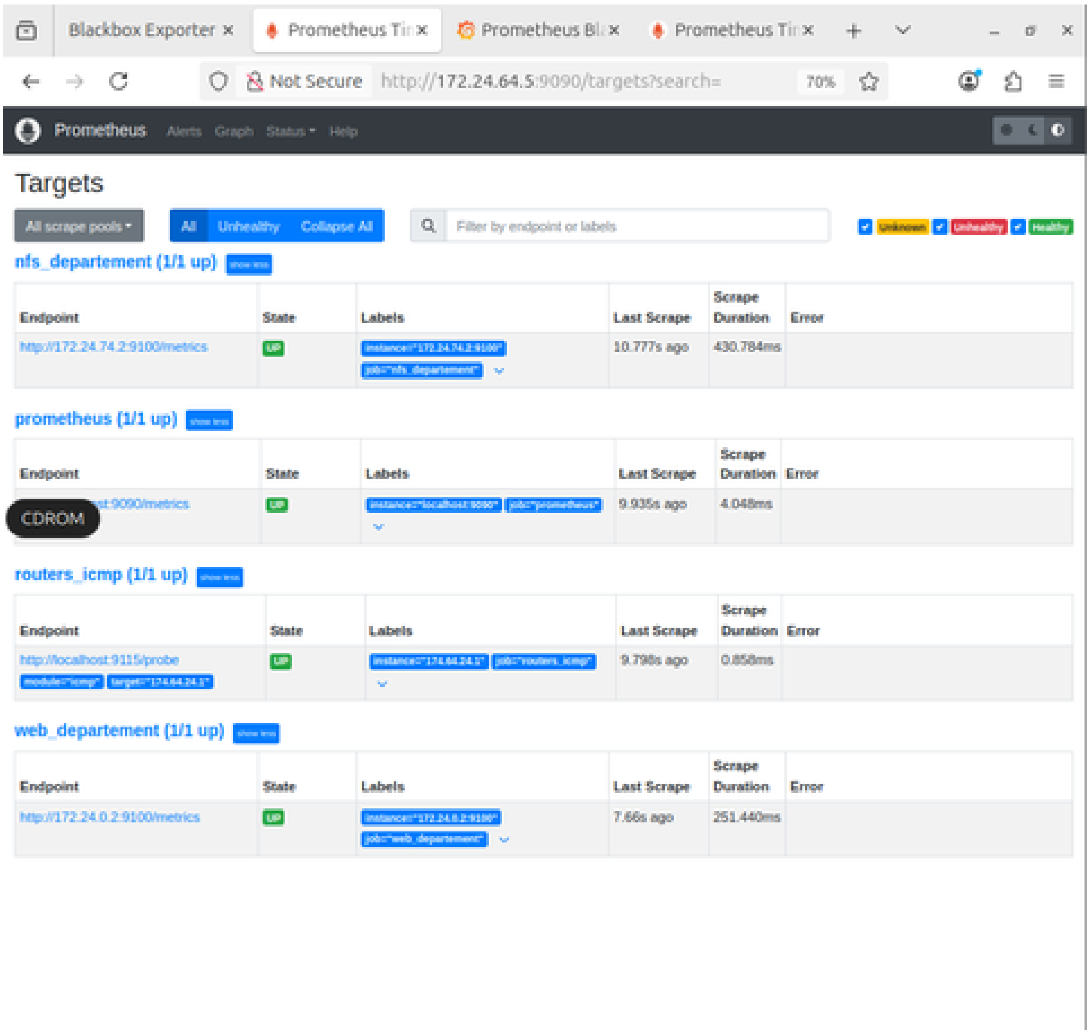


Figure 1: Prometheus Web Interface Showing Active Targets

10.5.2 Grafana Dashboard

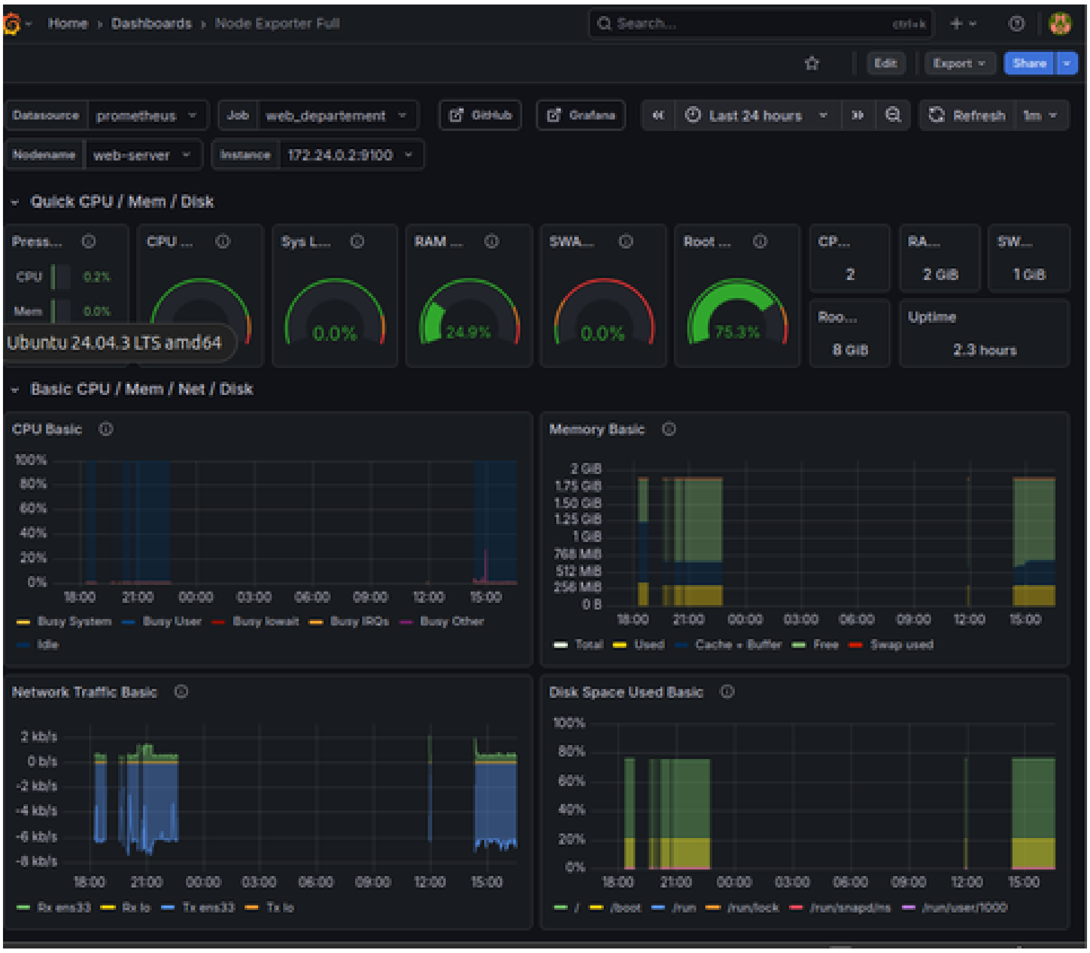


Figure 2: Grafana Node Exporter Dashboard (ID: 1860)

11 Conclusion

This document summarizes the complete network architecture for Tech Solutions (SARL). The simulated environment provides a realistic, scalable, and secure platform for training, experimentation, and enterprise design validation.

End of Report